

AI Ethics Copilot for THOA

A Proposed Rule-Based AI Screening System for Organ Transplant Documentation

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The Context: THOA Compliance

What is the THOA?

The Transplantation of Human Organs Act (THOA) was enacted to regulate organ donation and prevent commercial organ trafficking in India.

The Burden of Proof

To legally approve a transplant, hospitals must prove the relationship between the donor and recipient is genuine and not financially coerced.

- **Near-Relatives:** Require basic verification (Form 1).
- **Non-Relatives:** Require intense scrutiny, financial affidavits, and police clearance (Form 3, Form 11).

The Problem Statement

The Compliance Bottleneck

Every transplant case generates a dossier of 15 to 20 complex legal documents.

1. **Manual Verification is Slow:** Hospital Authorization Committees spend hours manually reading affidavits and verifying forms.
2. **High Risk of Human Error:** Overlooking a missing Embassy NOC (Form 21) or a mismatched date can result in severe legal penalties for the hospital.
3. **Administrative Delays:** The manual paperwork bottleneck delays urgent, life-saving surgeries.

Literature Review (1/3)

Domain: Legal Framework & Loopholes

Organ Trade in India: A Critical Analysis of the THOTA, 1994 — Merin Mathew (2025)

- **Finding:** Traces the evolution of the legal framework and the massive gaps driving illicit organ trade, noting that manual verification allows loopholes.
- **Relevance:** Proves the critical need for a strict, automated, and unbiased compliance checker that doesn't bend to hospital pressure.

Literature Review (2/3)

Domain: OCR in Healthcare Documentation

Deep Learning-Based NLP Data Pipeline for EHR — JAMIA Open (2023)

- **Finding:** Evaluated OCR and NLP pipelines specifically for scanned medical documents. Found high accuracy, but noted that image preprocessing is the most critical step for success.
- **Relevance:** Validates our architectural choice to use an OpenCV preprocessing pipeline before passing documents to Tesseract OCR and spaCy.

Literature Review (3/3)

Domain: Human-in-the-Loop Ethics

What Are Humans Doing in the Loop? — American Journal of Bioethics (2024)

- **Finding:** Argues that subjective, contextual judgments in healthcare should never be reduced to an automated "AI score."
- **Relevance:** Directly supports our core design constraint: the AI will objectively verify the *presence* of documents, but the human committee must evaluate the *ethics* (e.g., assessing genuine affection).

Proposed Objectives

What we intend to build:

1. **Automated Extraction:** Develop a pipeline to extract critical fields (Aadhaar, PAN, Dates, Income) from raw PDF scans.
2. **Rule-Based Screening:** Build a deterministic engine based on THOA sections to automatically flag anomalies.
3. **Explainability Layer:** Ensure every flagged anomaly provides a direct, traceable reference to a specific legal statute.
4. **Advisory Dashboard:** Create a UI for the Authorization Committee that highlights missing data in seconds.

Proposed Methodology: OCR Pipeline

Converting noisy PDFs into structured data:

1. **Rasterization:** Convert PDF uploads into high-resolution images (`pdf2image`).
2. **Pre-processing:** Apply Gaussian blur and Otsu thresholding via OpenCV to clean up poor-quality hospital scans.
3. **Text Extraction:** Utilize Tesseract OCR to pull raw text.
4. **Information Retrieval:**
 - Use **Regular Expressions** to extract structured formats like Aadhaar and PAN.
 - Use **spaCy NLP** to extract Names and Locations from unstructured legal affidavits.

Proposed Methodology: Rule Engine

Why deterministic rules instead of an LLM?

Legal compliance requires 100% accuracy, exact traceability, and zero hallucinations. An LLM is not suitable for legal gating.

The Engine:

- We will code 16 discrete Python rules mapped directly to THOA laws.
- **Example Rule:** If `relationship == 'NON_RELATIVE'`, system checks if `income_disparity_ratio > 10x`.
- **Outputs:** The engine will output predictable states: `PENDING_REVIEW` (Clean), `ANOMALIES` (Warnings), or `INCOMPLETE` (Missing Docs).

Proposed System Architecture

A Modern, Asynchronous Tech Stack

- **Frontend Interface:** React dashboard for hospital committees.
- **API Gateway:** FastAPI (Python) for strict data validation (Pydantic).
- **Background Processing:** Redis + Celery.
 - *Why?* OCR is computationally heavy; async processing prevents server timeouts during large uploads.
- **Database:** PostgreSQL for persistent, relational storage of case files.

Data Privacy & DPDP Compliance

Handling sensitive medical and financial data:

The Digital Personal Data Protection (DPDP) Act imposes strict rules on health data. Our system addresses this by:

1. **Encryption at Rest:** All Personal Identifiable Information (PII) like Aadhaar numbers will be encrypted using **AES-256-GCM** before database insertion.
2. **Data Masking:** Dashboards will automatically mask Aadhaar numbers (e.g., `XXXX-XXXX-1234`) for non-admin users.
3. **Immutable Auditing:** The database will track exactly who viewed or approved a case, ensuring total accountability.

Expected Challenges

1. The Handwriting Hurdle

OCR algorithms struggle significantly with handwritten regional languages (e.g., Hindi, Marathi) which are common in rural Indian affidavits. This requires a robust manual-override fallback.

2. The Forgery Gap

Our system can verify that Form 21 *exists* in the PDF. It cannot cryptographically verify if the document is a sophisticated forgery.

3. Automation Bias

The psychological risk that a busy Authorization Committee might simply "rubber-stamp" any case the AI marks as clean, ignoring subtle ethical red flags.

Evaluation Plan

How we will measure success:

1. **Synthetic Dataset:** We will construct a dataset of 100+ simulated transplant cases, including edge cases (e.g., underage donors, missing NOCs).
2. **Metric - OCR Accuracy:** Measure the percentage of correctly extracted critical fields.
3. **Metric - Rule Precision:** Ensure the engine flags 100% of illegal cases with zero false positives.
4. **Final Deliverable:** A working prototype demonstrating the end-to-end flow from PDF upload to an explainable audit report.

Conclusion

| **"A high-speed compliance filter."**

By automating the tedious, deterministic checks of the THOA documentation process, we aim to eliminate administrative bottlenecks and reduce legal risk.

The ultimate goal:

The software evaluates the paperwork, so the human Authorization Committee can dedicate 100% of their time to evaluating the ethics.

Thank You

Questions & Answers

Appendix: References (1/5)

Group 1: THOA & Organ Transplant Law in India

- *Organ Trade in India: A Critical Analysis of the THOTA, 1994* — Merin Mathew (IJCRT, 2025)
- *From Brain Death to Bureaucratic Delay: A Critical Appraisal of India's Organ Transplant Regime* (LiveLaw, 2026)
- *The Illicit Organ Trade: Biographical, Anatomical, Economic and Legal Aspects* (2025)
- *Organ Transplantation Law in India* (LawBhoomi, 2026)
- *Regulation of Organ Trafficking in India: A Critical Evaluation* (Legal Service India, 2026)
- *Analysis of Kidney and Liver Exchange Transplantation in India (2000–2025)* — Kute

Appendix: References (2/5)

Group 2: Data Privacy Law

- *Examining the Significance of the DPDP Act, 2023 in the Context of the Healthcare Industry* (Discover Public Health, 2025)

Group 3: Explainable AI in Regulatory/Legal Compliance

- *Ensuring Explainability in AI Systems for Financial Regulatory Compliance —* Santhosh Chitraju (SSRN, 2025)
- *Computational Compliance for AI Regulation: Blueprint for a New Research Domain* (arXiv, 2026)
- *Argumentation-Based Explainability for Legal AI: Comparative and Regulatory Perspectives* (arXiv, 2025)

Appendix: References (3/5)

Group 4: OCR & Document Information Extraction

- *Deep Learning-Based NLP Data Pipeline for EHR-Scanned Document Information Extraction* (JAMIA Open, 2022)
- *LayoutLMv3: Pre-training for Document AI with Unified Text and Image Masking* (arXiv, 2022)
- *Information Extraction from Unstructured Data Using Augmented Intelligence and Computer Vision* (arXiv, 2023)
- *BERT-Based NER and Relation Extraction for Electronic Medical Records* (Frontiers in Neuroscience, 2023)

Appendix: References (4/5)

Group 5: Human-in-the-Loop AI & Ethics in High-Stakes Decisions

- *Human-in-the-Loop Artificial Intelligence in Healthcare: Applications, Outcomes, and Implementation Challenges* (ScienceDirect, 2026)
- *Humans in the Loop, Lives on the Line: AI in High-Risk Decision-Making* (EA Journals, 2025)
- *What Are Humans Doing in the Loop? Co-Reasoning and Practical Judgment...* (American Journal of Bioethics, 2024)

Appendix: References (5/5)

Group 6: Hybrid Rule-Based + ML Systems for Compliance/Fraud

- *Advanced Fraud Detection Using Machine Learning Models: Enhancing Financial Transaction Security* (arXiv, 2025)
- *A Survey on Explainable Anomaly Detection* (arXiv, 2022)

Group 7: Swap/Paired Transplant Matching & Ethics

- *Adaptation, Comparison and Practical Implementation of Fairness Schemes in Kidney Exchange Programs* (arXiv, 2022)